Frequentist inference with weakly dependent data

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Ergodicity

Definition

A strictly stationary univariate time series $\{y_t\}$ is **ergodic for the mean** if $E|y_t| < \infty$ and

$$\hat{\mu}_y \equiv T^{-1} \sum_{t=1}^{T} y_t \overset{a.s.}{\to} E(y_t) \quad \text{as } T \to \infty.$$

- An ergodic process is one for which the (strong) LLN holds.
- It's possible to formulate weak low-level measure-theoretic conditions on $\{y_t\}$ that guarantee ergodicity for the mean. See Davidson (1994), ch. 13.4. But we will not focus on these conditions.
- Not every stationary time series with finite mean is ergodic. Example: $y_t = A$ for all $t \in \mathbb{Z}$, where $A \sim N(0, 1)$.
- Ergodicity requires the process to gradually "average out the past".

Mean-square convergence of sample mean

Proposition (Brockwell & Davis, 1991, Thm. 7.1.1)

If $\{y_t\}$ is univariate cov. stat., and $\Gamma_y(\ell) \to 0$ as $\ell \to \infty$, then

$$E[(\hat{\mu}_y - \mu_y)^2] \to 0 \quad \text{as } T \to \infty.$$

- This implies the weak LLN: $\hat{\mu}_y \xrightarrow{p} \mu_y$ as $T \to \infty$ (Chebyshev).
- Proof: $E(\hat{\mu}_y) = \mu_y$, and

$$\begin{aligned} T \operatorname{Var}(\hat{\mu}_y) &= T\, E\left[\left(T^{-1} \textstyle\sum_{t=1}^{T} (y_t - \mu_y)\right)^2\right] \\ &= T^{-1} \sum_{t=1}^{T} \sum_{s=1}^{T} \Gamma_y(t-s) = \sum_{|\ell|<T} \left(1 - \frac{|\ell|}{T}\right) \Gamma_y(\ell) \\ &\leq \sum_{|\ell|<T} |\Gamma_y(\ell)|. \end{aligned}$$

- Exercise: Finish the proof.

Variance of sample mean

- Consider an n-dimensional covariance stationary time series $\{y_t\}$.
- What is the variance of the sample mean $\hat{\mu}_y \equiv \frac{1}{T}\sum_{t=1}^{T} y_t$?
- Assume the ACF is abs. summable: $\sum_{\ell=-\infty}^{\infty} |\Gamma_{Y,ij}(\ell)| < \infty$.
- Then, by our previous calculations and the dominated convergence theorem,

$$T\,\mathrm{Var}(\hat{\mu}_y) = \sum_{|\ell|<T} \left(1 - \frac{|\ell|}{T}\right) \Gamma_y(\ell) \to \sum_{\ell=-\infty}^{\infty} \Gamma_y(\ell) \quad \text{as } T \to \infty.$$

Long-run variance

- Define the **long-run variance** of $\{y_t\}$:

$$\text{LRV}(y_t) \equiv \sum_{\ell=-\infty}^{\infty} \Gamma_y(\ell).$$

- Previous slide:

$$\text{Var}(\hat{\mu}_y) \approx \frac{1}{T}\text{LRV}(y_t).$$

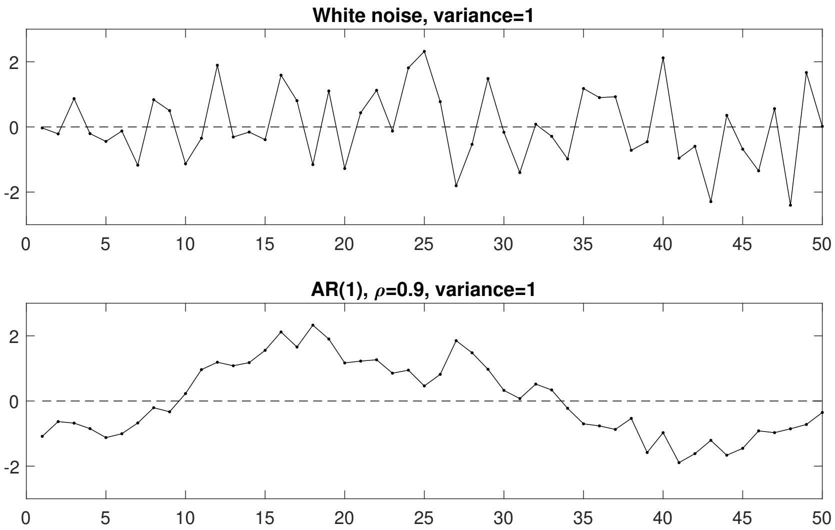
  This is $O(T^{-1})$ as in the case of i.i.d. data, but asy. var. is different.

- It is harder to estimate the mean of a positively serially correlated process (LRV $>$ Var) than that of a white noise process with the same variance.

- Exercise: Show that for a stationary AR(1) process $y_t = \rho y_{t-1} + u_t$, $u_t \sim WN(0, \sigma^2)$, we have

$$\text{LRV}(y_t) = \sigma^2/(1-\rho)^2 \neq \sigma^2/(1-\rho^2) = \text{Var}(y_t).$$

Long-run variance (cont.)



Long-run variance and spectral density

- Recall the definition of the spectral density of $\{y_t\}$:

$$s_y(\omega) \equiv \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} e^{-i\omega\ell} \Gamma_y(\ell), \quad \omega \in [-\pi, \pi].$$

- By inspection,

$$\mathrm{LRV}(y_t) = 2\pi s_y(0).$$

- The long-run variance depends on the low-frequency behavior of the time series (in fact, the lowest frequency).
- We already know how to estimate the spectrum, and we will soon discuss LRV estimation in more detail. But for now, just assume we have a consistent estimator.

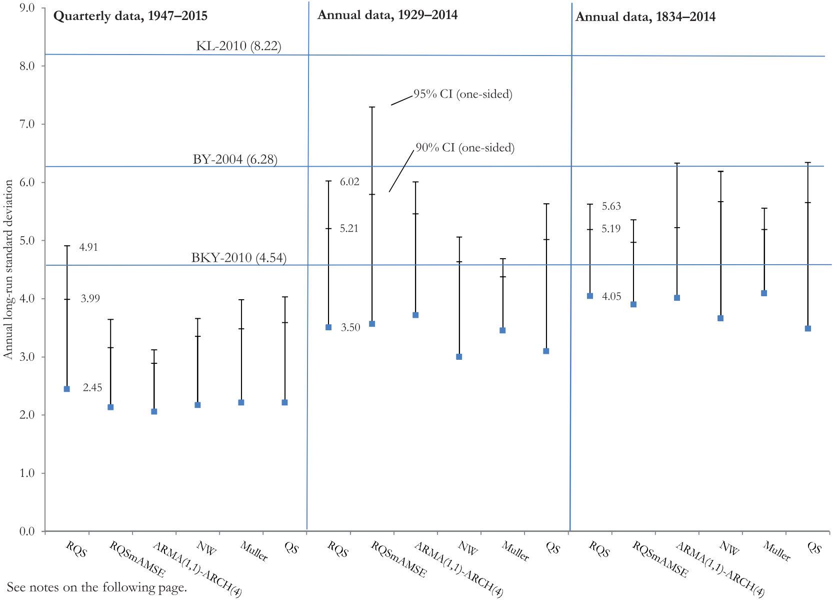
Application: Long-run riskiness of consumption growth

- In some (rare) cases, LRVs are of intrinsic economic interest.
- The **long-run risk** model assumes that asset prices are determined by the correlation of returns with consumption growth rates (and risk-free interest rates) into the far future.
- Contrasts with the classic Lucas-Mehra-Prescott consumption asset pricing model, where only the contemporaneous covariance of returns with consumption growth Δc_{t+1} matters.
- Dew-Becker (2017) shows that the std. dev. of the stochastic discount factor (SDF) in the long-run risk model is given by

$$\text{std}(M_t) \approx \gamma \sqrt{\text{LRV}(\Delta c_{t+1})},$$

  in the limit as the intertemporal elasticity of substitution tends to 0. γ: coeff. of relative risk aversion in investors' Epstein-Zin preferences.
- In turn, the std. dev. of the SDF allows us to compare the model's predictions to the observed equity premium.

Application: Long-run riskiness of cons'n growth (cont.)



See notes on the following page.

Source: Dew-Becker (2017)

CLT for linear process

- Consider the **linear process** (MA(∞) with i.i.d. innovations)

$$y_t = \mu + \Theta(L)u_t, \quad u_t \overset{i.i.d.}{\sim} WN(0, \sigma^2),$$

where $\Theta(L) = \sum_{\ell=0}^{\infty} \Theta_\ell L^\ell$ is **one-summable**: $\sum_{\ell=0}^{\infty} \ell |\Theta_\ell| < \infty$.

- We already know

$$E(\hat{\mu}_y) = \mu, \quad \text{Var}(\hat{\mu}_y) \approx T^{-1}\text{LRV}(y_t) = T^{-1}\sigma^2\Theta(1)^2.$$

- To get a central limit theorem (CLT), we apply the **Phillips-Solo device** (cf. Problem Set 1)

$$\Theta(L) = \Theta(1) + (1-L)B(L), \quad B(L) \equiv -\sum_{\ell=0}^{\infty}\left(\sum_{m=\ell+1}^{\infty}\Theta_m\right)L^\ell.$$

$B(L)$ is summable because $\Theta(L)$ is one-summable.

CLT for linear process (cont.)

- Using the Phillips-Solo device, we can write

$$y_t = \mu + \Theta(1)u_t + \eta_t - \eta_{t-1}, \quad \eta_t \equiv B(L)u_t.$$

- Then

$$\hat{\mu}_y \equiv T^{-1}\textstyle\sum_{t=1}^{T} y_t = \mu + \Theta(1)\, T^{-1}\textstyle\sum_{t=1}^{T} u_t + T^{-1}(\eta_T - \eta_0).$$

- Thus, $\hat{\mu}_y$ inherits its limiting behavior from the i.i.d. series $\{u_t\}$,

$$\sqrt{T}(\hat{\mu}_y - \mu) = \Theta(1)\underbrace{T^{-1/2}\textstyle\sum_{t=1}^{T} u_t}_{\xrightarrow{d} N(0,\sigma^2)} + \underbrace{O_p(T^{-1/2})}_{\xrightarrow{p} 0} \xrightarrow{d} N(0, \sigma^2\Theta(1)^2),$$

  using the standard CLT and Slutsky's lemma.

- One-summability of $\Theta(L)$ is stronger than needed for the CLT (cf. B&D, ch. 7.1). But the above argument is transparent and flexible.

CLT for martingale difference sequence

- $\{y_t\}$ is a **martingale difference sequence** (MDS) wrt. the **filtration** (increasing sequence of sigma algebras) $\mathcal{F}_t$ if:

  1. y_t is measurable wrt. $\mathcal{F}_t$ for all t. E.g., $\mathcal{F}_t = \sigma(\{y_\tau\}_{\tau \leq t})$.
  2. $E(y_t \mid \mathcal{F}_{t-1}) = 0$ for all t.

- Weaker condition than $\{y_t\}$ being i.i.d. with $\mu_y = 0$. Allows for time-varying conditional volatility.

Proposition (Davidson, 1994, Thm. 23.16 & 24.3)

Let $\{y_t\}$ be a univariate strictly stationary MDS wrt. some filtration $\mathcal{F}_t$, and assume $\sigma_y^2 \equiv \text{Var}(y_t) < \infty$. If

$$T^{-1} \sum_{t=1}^{T} y_t^2 \xrightarrow{p} \sigma_y^2 \quad \text{as } T \to \infty,$$

then $\sqrt{T}\hat{\mu}_y \xrightarrow{d} N(0, \sigma_y^2)$ as $T \to \infty$.

Gordin's CLT

- Although the MDS condition is weaker than i.i.d.-ness, it still implies no serial correlation. Here is a CLT that allows for weak dependence.

Proposition (Hayashi, 2000, equation 6.5.8)

Let $\{y_t\}$ be an n-dim. ergodic and strictly stationary time series. Let $\|\cdot\|$ denote the Euclidean norm. Assume:

i) $E(\|y_t\|^2) < \infty$.

ii) $E(\|E(y_t \mid \{y_\tau\}_{-\infty<\tau\leq t-j})\|^2) \to 0$ as $j \to \infty$.

iii) $\sum_{j=0}^{\infty} \sqrt{E(\|r_{t,j}\|^2)} < \infty$, where $r_{t,j} \equiv E(y_t \mid \{y_\tau\}_{-\infty<\tau\leq t-j}) - E(y_t \mid \{y_\tau\}_{-\infty<\tau\leq t-j-1})$.

Then $E(y_t) = 0_{n\times 1}$, the ACF of $\{y_t\}$ is abs. summ., and

$$\sqrt{T}\hat{\mu}_y \xrightarrow{d} N(0_{n\times 1}, \mathrm{LRV}(y_t)) \quad \text{as } T \to \infty.$$

Example: time series regression

- Consider a time series linear regression model:

$$y_t = \beta_0' x_t + \varepsilon_t, \quad E(x_t \varepsilon_t) = 0_{k \times 1}.$$

  The processes $\{x_t, \varepsilon_t\}$ may be serially correlated.

- Asymptotics for OLS estimator:

$$\sqrt{T}(\hat{\beta} - \beta_0) = \left(\frac{1}{T} \sum_{t=1}^{T} x_t x_t' \right)^{-1} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^{T} x_t \varepsilon_t \right)$$
$$\xrightarrow{d} N\Big(0, S_x^{-1} \Omega S_x^{-1}\Big), \quad S_x \equiv E(x_t x_t'), \ \Omega \equiv \text{LRV}(x_t \varepsilon_t).$$

Application: Testing inflation forecast rationality

	Michigan	*Michigan-experimental*	*Livingston*	*SPF (GDP deflator)*
Panel A: testing for bias: $\pi_t - E_{t-12}\,\pi_t = \alpha$				
α: mean error	0.42%	–0.09%	0.63%**	–0.02%
(Constant only)	(0.29)	(0.34)	(0.30)	(0.29)
Panel B: Is information in the forecast fully exploited? $\pi_t - E_{t-12}\,\pi_t = \alpha + \beta\, E_{t-12}\pi_t$				
β: $E_{t-12}\,[\pi_t]$	0.349**	–0.060	0.011	0.026
	(.161)	(.207)	(.142)	(.128)
α: constant	–1.016%*	–0.182%	0.595%	–0.132%
	(.534)	(.721)	(.371)	(.530)
Adj. R^2	0.197	–0.003	–0.011	–0.007
Reject eff.? $\alpha = \beta = 0$	Yes	No	Yes	No
(p-value)	(p = 0.088)	(p = 0.956)	(p = 0.028)	(p = 0.969)
Panel C: Are forecasting errors persistent? $\pi_t - E_{t-12}\,\pi_t = \alpha + \beta\,(\pi_{t-12} - E_{t-24}\pi_{t-12})$				
β: $\pi_{t-12} - E_{t-24}\,[\pi_{t-12}]$	0.371**	.580***	0.490***	0.640***
	(0.158)	(0.115)	(0.132)	(0.224)
α: constant	0.096%	0.005%	0.302%	–0.032%
	(0.183)	(0.239)	(0.210)	(0.223)
Adj. R^2	0.164	0.334	0.231	0.375

Source: Mankiw, Reis & Wolfers (2004)

Application: Testing inflation forecast rationality (cont.)

Panel D: Are macroeconomic data fully exploited? $\pi_t - E_{t-12}\pi_t = \alpha + \beta\ E_{t-12}\ [\pi_t] + \gamma\,\pi_{t-13} + \kappa\, i_{t-13} + \delta\, U_{t-13}$

α: constant	–0.816%	0.242%	4.424%***	3.566%***
	(0.975)	(1.143)	(0.985)	(0.970)
β: $E_{t-12}\ [\pi_t]$	0.801***	–0.554***	0.295	0.287
	(0.257)	(0.165)	(0.283)	(0.308)
γ: inflation$_{t-13}$	–0.218*	0.610***	0.205	0.200
	(0.121)	(0.106)	(0.145)	(0.190)
κ: Treasury bill$_{t-13}$	–0.165**	–0.024	–0.319***	–0.321***
	(0.085)	(0.102)	(0.106)	(0.079)
δ: unemployment$_{t-13}$	0.017	–0.063	–0.675***	–0.593***
	(0.126)	(0.156)	(0.175)	(0.150)
Reject eff.? $\gamma = \kappa = \delta = 0$	Yes	Yes	Yes	Yes
(p-value)	(p = 0.049)	(p = 0.000)	(p = 0.000)	(p = 0.000)
Adjusted R^2	0.293	0.382	0.306	0.407
Sample	Nov. 1974–May 2002	1954, Q4–2002, Q1	1954, H1–2001, H2	1969, Q4–2002, Q1
Periodicity	Monthly	Quarterly	Semiannual	Quarterly
N	290	169	96	125

1. ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels, respectively (Newey-West standard errors in parentheses; correcting for autocorrelation up to one year).

Source: Mankiw, Reis & Wolfers (2004)

Mixing processes

- Consider a stationary process $\{y_t\}$ and define $\mathcal{F}_\ell^m \equiv \sigma(\{y_\tau\}_{\ell \leq \tau \leq m})$. The process is said to be **strongly mixing** if

$$\alpha_j \equiv \sup_t \sup_{G \in \mathcal{F}_{-\infty}^t, H \in \mathcal{F}_{t+j}^\infty} |P(G \cap H) - P(G)P(H)| \to 0 \quad \text{as } j \to \infty.$$

  Intuitively, we have "independence in the limit".

- There exist various LLNs and CLTs for strongly mixing processes. These typically require $\alpha_j = O(j^{-\varphi})$ at some appropriate rate $\varphi > 0$. See Davidson (1994), ch. 14, 19–20, 24.

- Mixing is a strong requirement. For example, the following linear process is not strongly mixing (see Davidson, 1994, Thm. 14.7):

$$y_t = \tfrac{1}{2} y_{t-1} + z_t, \quad z_t \overset{iid}{\sim} \text{Bernoulli}(\tfrac{1}{2}).$$

- Strong mixing of $\{y_t\}$ is too much to ask for, since we don't need independence for *all* kinds of events G, H to prove the LLN/CLT.

Near-epoch dependence

- One way to weaken the mixing requirement is to consider transformations of mixing variables.
- A strictly stationary time series $\{y_t\}$ is said to be "**near-epoch dependent in mean-square norm on** $\{z_t\}$" if

$$\nu_j \equiv \sqrt{E\left[\left\{y_t - E(y_t \mid \{z_\tau\}_{t-j\le\tau\le t+j})\right\}^2\right]} \to 0 \quad \text{as } j \to \infty.$$

- Hence, y_t is well approximated by some function of $\{z_\tau\}_{t-j\le\tau\le t+j}$, for large j.
- If $\{y_t\}$ is itself mixing, and $\nu_j = O(j^{-\varphi})$ at an appropriate rate $\varphi > 0$, then one can establish LLNs/CLTs for $\{y_t\}$. See Davidson (1994), ch. 17, 19.4, 20.6, 24.4.

Generalized Method of Moments

- Consider GMM with moment conditions

$$E[g(y_t, \theta_0)] = 0_{r \times 1}.$$

- GMM estimator:

$$\hat{\theta} \equiv \text{argmin}_{\theta \in \Theta} \left(T^{-1} \textstyle\sum_{t=1}^{T} g(y_t, \theta) \right)' \hat{W} \left(T^{-1} \textstyle\sum_{t=1}^{T} g(y_t, \theta) \right).$$

- From your first-year courses:

$$\hat{\theta} - \theta_0 = (G'WG)^{-1} G'W \left(\frac{1}{T} \sum_{t=1}^{T} g(y_t, \theta_0) \right) + o_p(T^{-1/2})$$

$$\Longrightarrow \sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N\Big(0, (G'WG)^{-1} G'W\Omega WG(G'WG)^{-1}\Big),$$

$$G \equiv E\left[\frac{\partial g(y_t, \theta_0)}{\partial \theta'}\right], \quad \Omega \equiv \text{LRV}(g(y_t, \theta_0)).$$

- Special case: 2SLS.

Application: New Keynesian Phillips Curve

- **New Keynesian Phillips Curve** relates inflation π_t to inflation expectations, economic slack x_t, and a cost-push shock ν_t:

$$\pi_t = \gamma_f E_t(\pi_{t+1}) + \gamma_b \pi_{t-1} + \lambda x_t + \nu_t.$$

- γ_f: degree of forward-looking price setting.
- γ_b: degree of backward-looking price setting.
- λ: slope of Phillips Curve.
- Assume cost-push shock is unpredictable, $E_{t-1}(\nu_t) = 0$.
- Parameter values matter hugely for optimal monetary policy.
- Issue: We arguably do not observe firms' inflation expectations $E_t(\pi_{t+1})$. May differ from household/professional survey forecasts.

Application: New Keynesian Phillips Curve (cont.)

- Roberts (1995), Galí & Gertler (1999): Define forecast error $\varepsilon_{t+1} \equiv \pi_{t+1} - E_t(\pi_{t+1})$. Can then substitute out expectations:

$$\pi_t = \gamma_f \pi_{t+1} + \gamma_b \pi_{t-1} + \lambda x_t + u_{t+1},$$

  where $u_{t+1} \equiv \nu_t - \gamma_f \varepsilon_{t+1}$.

- Assume **rational expectations**: $E_t(\varepsilon_{t+1}) = 0$. Then $E_{t-1}(u_{t+1}) = 0$.

- GMM moment conditions (IV):

$$E\left[(\pi_t - \gamma_f \pi_{t+1} - \gamma_b \pi_{t-1} - \lambda x_t) Z_{t-1}\right] = 0_{r \times 1},$$

  where Z_{t-1} is an r-dim. vector of variables known at time $t-1$.

- Hundreds of papers have estimated versions of these moment equations, following Galí & Gertler (1999). $u_{t+1} Z_{t-1}$ could be serially correlated, so need to estimate LRV sandwich matrix.
- Papers differ slightly in choice of instruments Z_{t-1}, inflation measure, slack measure, no. of inflation lags, time period, etc.

Application: New Keynesian Phillips Curve (cont.)

TABLE 3
BASELINE GIV ESTIMATES USING DIFFERENT DATA VINTAGES

Data vintage	Const.	λ	γ_f	γ_b	Hansen test
1998	0.041	0.026	0.615	0.340	5.263
	(0.030)	(0.013)	(0.057)	(0.058)	[0.628]
2012	–0.049	0.018	0.719	0.240	9.816
	(0.040)	(0.012)	(0.099)	(0.095)	[0.199]

Notes: Comparison of GIV estimates of the hybrid NKPC based on 1998 and 2012 vintages of data. The estimation sample is 1970q1 to 1998q1. Inflation: GDP deflator. Labor share: NFB. Instruments: four lags of inflation and two lags of the labor share, wage inflation, and quadratically-detrended output. Estimation method: CUE GMM. Weight matrix: Newey and West (1987) with automatic lag truncation (4 lags). Standard errors in parentheses and p-values in square brackets.

Source: Mavroeidis et al. (2014)

Application: New Keynesian Phillips Curve (cont.)

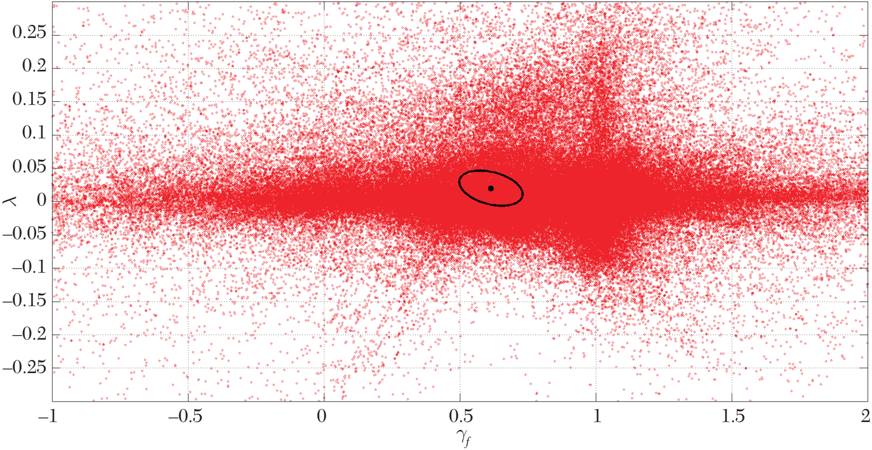


Figure 4. Point Estimates: Labor Share Specifications

Notes: Point estimates of λ, γ_f from the various specifications listed in table 4 that use the labor share as forcing variable, excluding real-time and survey instrument sets. The black dot and ellipse represent the point estimate and 90 percent joint Wald condence set from the 1998 vintage results in table 3.

Source: Mavroeidis et al. (2014)

Maximum likelihood

- Maximum likelihood estimator (MLE):

$$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmax}} \sum_{t=1}^{T} \log p(y_t \mid \{y_s\}_{s<t}, \theta).$$

- Corresponds to just-identified GMM with moment function (called the **score function**)

$$g(y_t, \theta) = \frac{\partial \log p(y_t \mid \{y_s\}_{s<t}, \theta)}{\partial \theta}.$$

- If model is correctly specified, one can show that $g(y_t, \theta_0)$ is a MDS, so $\Omega = \text{Var}(g(y_t, \theta_0))$. Moreover, $\Omega = -G$, and thus $\text{asyVar}(\hat{\theta}) = \Omega^{-1} = -G^{-1}$.

- But if we want our standard errors to be robust to misspecification, we need to estimate the LRV of the score process.

Minimum distance

- Suppose we have a structural model with parameters $\theta_0 \in \Theta \subset \mathbb{R}^k$.
- The model implies $\eta_0 = h(\theta)$ iff $\theta = \theta_0$, where $h\colon \Theta \to \mathbb{R}^r$ is a known function and η_0 are some "reduced-form" parameters.
- We have access to some estimator $\hat{\eta}$ satisfying

$$\sqrt{T}(\hat{\eta} - \eta_0) \xrightarrow{d} N(0_{r\times 1}, \Psi)$$

  for some $r \times r$ LRV matrix Ψ.
- Examples of η_0: moments, regression coefficients.
- Minimum distance estimator of θ_0 (formal version of **calibration**):

$$\hat{\theta} \equiv \underset{\theta\in\Theta}{\operatorname{argmin}} \{\hat{\eta} - h(\theta)\}'\hat{W}\{\hat{\eta} - h(\theta)\}.$$

Minimum distance (cont.)

- FOC:
$$\frac{\partial h(\hat{\theta})'}{\partial \theta} \hat{W} \{ \hat{\eta} - h(\hat{\theta}) \} = 0_{k \times 1}.$$
- Assume $\hat{W} \overset{p}{\to} W$, $h(\cdot)$ is ctsly diff'ble, and $D \equiv \frac{\partial h(\theta_0)}{\partial \theta'}$ has full rank. Then a Taylor expansion of the FOC gives
$$\begin{aligned} \sqrt{T}(\hat{\theta} - \theta_0) &= (D'WD)^{-1} D'W\sqrt{T}(\hat{\eta} - \eta_0) + o_p(1) \\ &\overset{d}{\to} N\left(0_{k \times 1}, (D'WD)^{-1} D'W\Psi WD(D'WD)^{-1}\right). \end{aligned}$$
- Efficient weight matrix: $W_{\text{eff}} \equiv \Psi^{-1}$. Yields asy. var. $(D'\Psi^{-1}D)^{-1}$.
- In practice, often use model simulations to compute $h(\cdot)$. To compute s.e. for $\hat{\theta}$, just need $\hat{\Psi}$ and $\hat{D} = \frac{\partial h(\hat{\theta})}{\partial \theta'}$ (e.g., finite differences).
- Technical details: Newey & McFadden (1994).